

ANALOG-TO-DIGITAL CONVERSION

ADC

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

ADC Process

Analog Input Demo

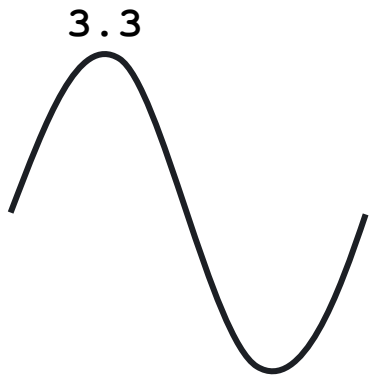


ANALOG-TO-DIGITAL CONVERSION

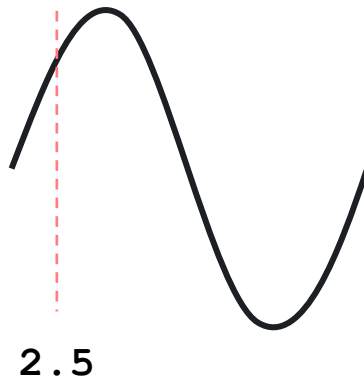


ADC PROCESS

Analog Voltage



Sampling



Quantization

A 12-bit ADC can represent $2^{12} = 4096$ different levels, ranging from 0 to 4095.

$$\text{ADC} = \frac{v_{in}}{v_{ref}} \cdot (2^n - 1)$$

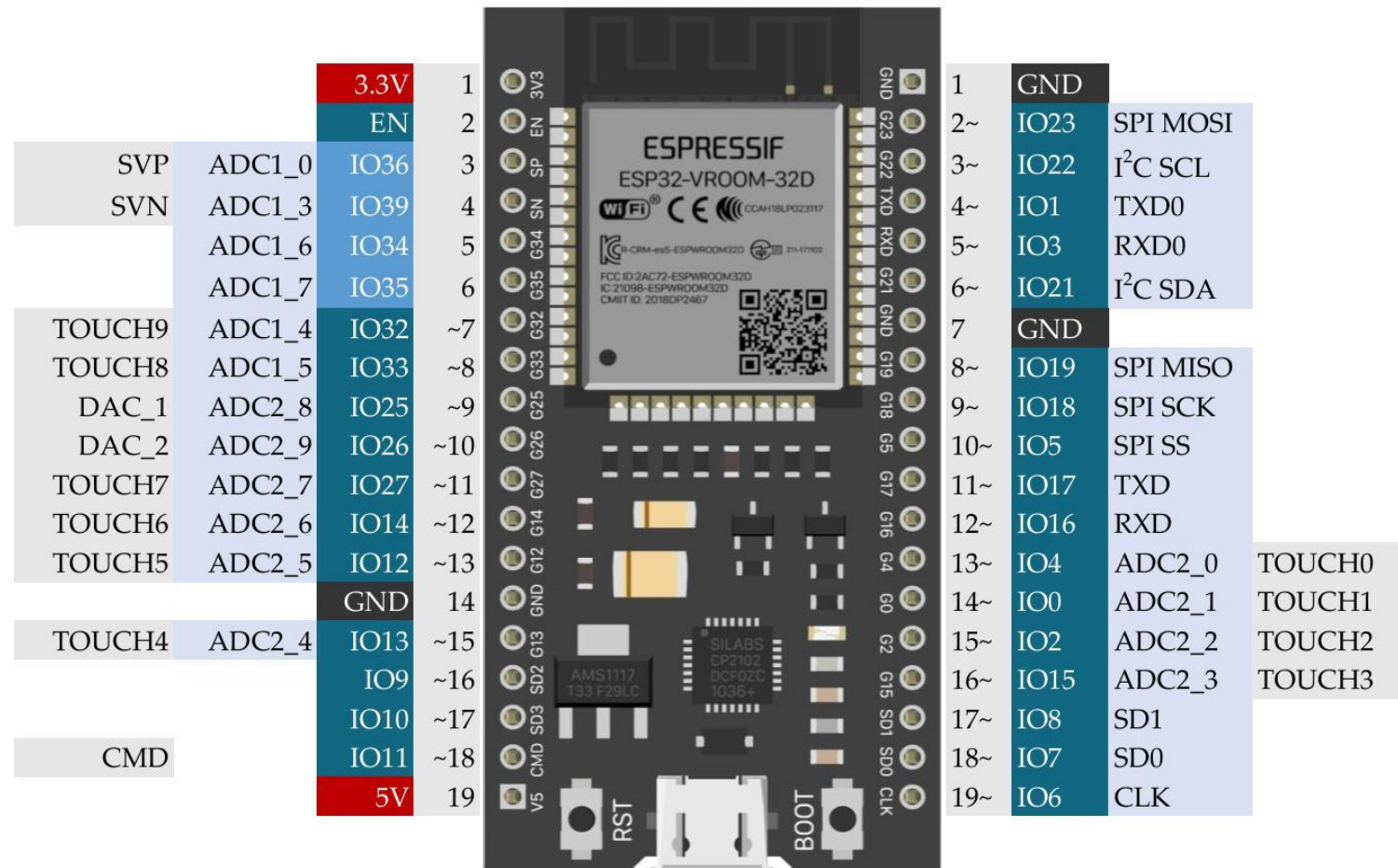
$$\text{ADC} = \frac{2.5}{3.3} \cdot 4095$$

Digital Value

$$\text{ADC} = 3102$$



DEVKITC GPIOs MAP

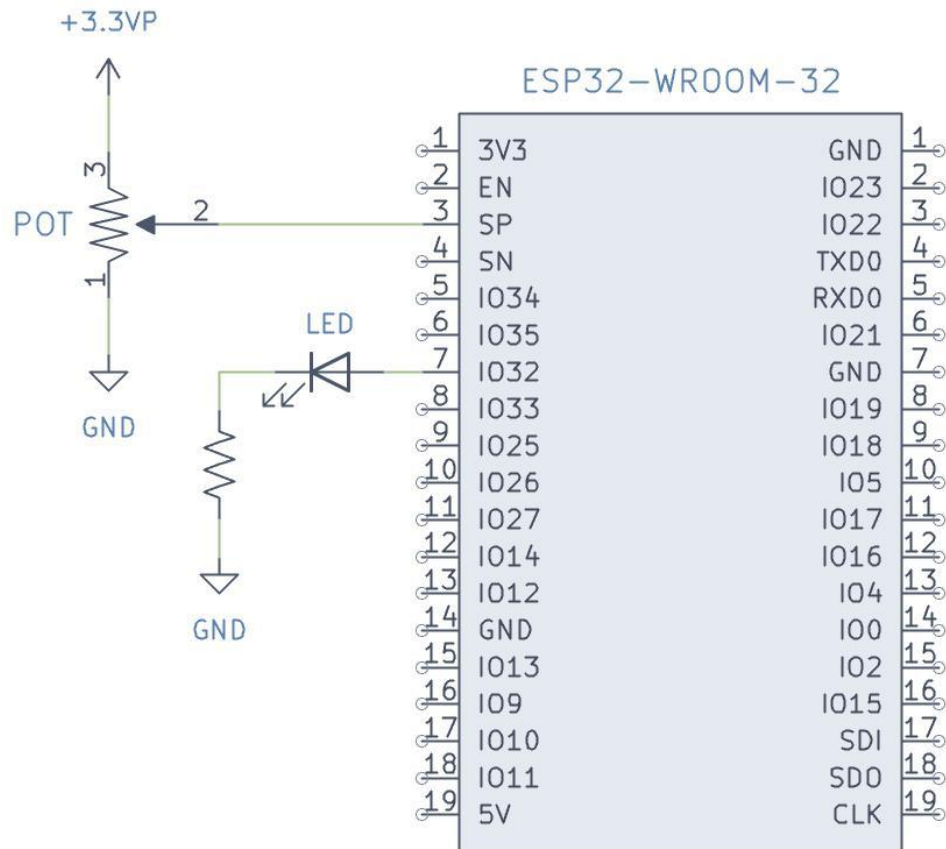


GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

[ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation](#)

ANALOG INPUT DEMO



`analogRead()` – returns a **numeric** value proportional to the input voltage.



LABORATORY

